

# Project Proposal

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2026 Spring

## 1. What problem am I solving? State it clearly.

Investigating causal drivers of neighborhood opposition (NIMBYism) to housing [Einstein et al., 2019, p. 28]. **Goal:** separate stable, **invariant** predictors of resistance from **spurious associations** driven by **environment-specific confounders** [Peters et al., 2018, p. 84]. **Aim:** find features predicting opposition consistently across the distinct **market regimes** of the 2018–2025 period.

## 2. Why is this problem important? To whom?

Critical for urban planners. Current models overfit to specific eras, failing when market regimes shift [Arjovsky et al., 2019, p. 1]. Identifying *invariant* causes enables policy design addressing fundamental concerns rather than symptoms of a moment, ensuring **robustness** [Bühlmann, 2020, p. 404].

## 3. What is my strategy for solving it?

Using **Invariant Risk Minimization (IRM)** [Arjovsky et al., 2019, p. 5] and **Bayesian Invariant Prediction (BIP)** [Wu et al., 2025b, p. 2]:

1. **Environments:** Partition data via **expanding window Cross-Validation** into **Training Environments** ( $\mathcal{E}_{tr}$ ) and **Test Environments** ( $\mathcal{E}_{te}$ ):

| Window | Training ( $\mathcal{E}_{tr}$ ) | Test ( $\mathcal{E}_{te}$ ) |
|--------|---------------------------------|-----------------------------|
| 1      | 2018                            | 2019–2025                   |
| 2      | 2018–2019                       | 2020–2025                   |
| ...    | ...                             | ...                         |
| $N$    | 2018–2024                       | 2025                        |

2. **Representation:** Extract features  $\Phi(X)$  from property data and protest letters [Schölkopf et al., 2021, p. 15].
3. **Methods:** We implement the two core paradigms discussed in the seminar:

- **IRM (Regularized Optimization):** Learn a representation  $\Phi$  such that a **fixed dummy classifier**  $w = 1.0$  is simultaneously optimal across all  $\mathcal{E}_{tr}$ , enforced via a gradient norm penalty [Arjovsky et al., 2019, p. 5]:

$$\min_{\Phi} \sum_{e \in \mathcal{E}_{tr}} R^e(\Phi) + \lambda \|\nabla_{w|w=1.0} R^e(w \cdot \Phi)\|^2$$

- **BIP (Posterior Inference):** Define a latent **feature selection vector**  $z \in \{0, 1\}^p$  and compute the **posterior**  $p(z|D)$  to identify the subset of features with a **stable predictive relationship** to the outcome [Wu et al., 2025b, p. 2].

## 4. Why is this solution good? Where does it fall short?

**It falls short because** it assumes an **invariant causal mechanism**. The invariance assumption fails if the underlying reasons for NIMBYism undergo **mechanism change** over time. For instance, neighborhood opposition may shift from traffic congestion fears to displacement concerns as demographics evolve.

## 5. How can I demonstrate that the solution works?

Train on expanding  $\mathcal{E}_{tr}$ , test on all **future horizons** (2019–2025) in  $\mathcal{E}_{te}$ . Successful Invariant Model demonstrates **predictive stability** through the **invariance of its AUC** score across all horizons, maintaining reliability where baselines fail under **distribution shift** [Krueger et al., 2021, p. 1].

## References

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